Posterior-Moment In-Context Solvers for Bayesian Near-Field Linear Sampling

Technical results note

4 August 2026

Abstract

We formulate the original two-dimensional near-field linear sampling method (LSM) as a kernel Bayesian inverse problem and adapt the parallel posterior- predictive recurrences of Kang, Lee and Cheng to this setting. Point sources illuminate one to six sound-soft obstacles and a distinct receiver array records the complex scattered near field. A prescribed von Mises/softmax covariance is a modelling choice: neither its nonlinearity nor a temperature is learned. A single tied loop solves both the probe equation, which determines the posterior mean, and the matrix equation required for posterior covariance. Richardson, heavy-ball (HB), Chebyshev and PCG cells therefore share the same encoder, decoder and Bayesian semantics. We prove the corresponding depth laws and show why learning only population whitening cannot remove obstacle-dependent eigendirections. A controlled multi-obstacle audit reports localization, residual, calibration and robustness. Its main negative result is useful: finite-horizon HB benefits from deliberately conservative effective endpoints, whereas plugging in the exact spectral endpoints produces a large transient. Small residuals require either substantially greater depth or adaptive Krylov coefficients.

1 Scope and correction of the problem

This note concerns deterministic active *near-field* LSM, not scalar one-dimensional Gaussian-process regression, not a Born far-field surrogate, and not the random-source LSM variants of Montanelli and coauthors [4, 5]. For each task, source and receiver locations lie on separate circles. The measured object is the full multistatic near-field matrix N_D generated by a sound-soft obstacle D. The random objects in training are obstacles, acquisition perturbations and additive noise; the incident point sources are controlled and deterministic.

2 Physical near-field model

Let $\Gamma_s = \{y_j\}_{j=1}^{n_s}$ and $\Gamma_r = \{x_i\}_{i=1}^{n_r}$ denote source and receiver curves. With $\Phi_k(x, y) = \frac{\mathrm{i}}{4} H_0^{(1)}(k|x-y|)$, the incident field from y_j is $u_j^i(x) = \Phi_k(x, y_j)$. For a sound-soft obstacle,

$$(\Delta + k^2)u_j^s = 0 \quad \text{outside } D, \qquad u_j^s = -u_j^i \quad \text{on } \partial D,$$

together with the Sommerfeld radiation condition. The data matrix is $(N_D)_{ij} = u_j^s(x_i)$. Numerical data are generated by boundary collocation with interior fundamental sources; the mean relative boundary residual in the audit is 1.35e-06. This is an independent sound-soft forward model, not reused code from `lsmlab`.

For every sampling point z in a two-dimensional grid, define $\phi_z = (\Phi_k(x_i, z))_i$. After whitening the known receiver-noise covariance R_Γ, we retain the same symbols for $R_\Gamma^{-1/2} N_D$ and $R_\Gamma^{-1/2}\phi_z$. This is the original near-field sampling equation $N_D g_z \simeq \phi_z$ in a statistically explicit form.

3 A fixed nonlinear feature model

For source angles θ_i, let

$$W_{ij} = \exp[\gamma \cos(\theta_i - \theta_j)], \qquad K_\Gamma = (1-\eta)I + \eta D_W^{-1/2} W D_W^{-1/2}.$$

This is the symmetric positive version of row-softmax attention. In the reported model $\gamma = 1.0$ and $\eta = 0.2$ are fixed before training. The softmax nonlinearity selects the GP feature space; it is not an optimizable temperature. Other prior knowledge can be encoded by replacing $\cos(\theta_i - \theta_j)$ with a task-appropriate feature similarity while preserving positive definiteness.

The proper-complex prior is $g \sim \mathcal{CN}(0, K_\Gamma)$ and whitened noise has identity covariance. Set

$$H_D = N_D K_\Gamma N_D^* + I.$$

For the matrix $\Phi = [\phi_z]_z$, the two posterior equations are

$$H_D Q_\mu = \Phi, \qquad H_D Q_\Sigma = N_D K_\Gamma. \tag{1}$$

Proposition 1 (Parallel posterior moments)**.** *The conditional coefficient field associated with the probe matrix has mean and covariance*

$$M_D = K_\Gamma N_D^* Q_\mu, \qquad \Sigma_D = K_\Gamma - K_\Gamma N_D^* Q_\Sigma.$$

Consequently, applying one tied linear recurrence to the concatenated right- hand side $[\Phi, N_D K_\Gamma]$ *computes both posterior moments with identical solver coefficients and depth.*

Proof. These are the Woodbury-form conditioning identities for a linear proper- complex Gaussian model. Both expressions contain the same inverse H_D^{-1}, which proves the shared recurrence statement. □

This is the exact near-field analogue of the two parallel recurrences used for posterior predictive mean and variance in Kang et al. [1]. It also clarifies terminology: the matrices $B^* A^j B$ used below are *spectral Krylov statistics* for the controller, not posterior moments.

4 Architecture

Let $C_{\mathrm{pop},\Gamma}$ be a geometry-conditioned factor diagonal in the prescribed receiver-feature basis. The task supplies

$$A_D = s_D^{-1} C_{\mathrm{pop},\Gamma}^* H_D C_{\mathrm{pop},\Gamma}, \qquad B_D = s_D^{-1} C_{\mathrm{pop},\Gamma}^* [\Phi, N_D K_\Gamma],$$

where the row bound s_D ensures $\lambda_{\max}(A_D) \leq 1$. A fixed random sketch compresses both right-hand-side blocks, and the controller reads

$$\mathcal{E}_D = \left\{ \frac{Z_0^* A_D^j Z_0}{\|Z_0\|_F^2} \right\}_{j=0}^{J}, \qquad J = 8,$$

together with the safe lower bound and population-gain summaries. The 0.73M- parameter MLP outputs two *effective* endpoints. The physical matrix, kernel, posterior formulas and recurrent solver are never learned. The same encoder and decoder are used by all stationary cells.

The output score is based on $q_z = g_z^* K_\Gamma^{-1} g_z$. For $g_z \sim \mathcal{CN}(m_z, \Sigma_D)$,

$$\begin{aligned} \mathbb{E} q_z &= m_z^* K_\Gamma^{-1} m_z + \operatorname{tr}(K_\Gamma^{-1} \Sigma_D), \\ \operatorname{Var} q_z &= \operatorname{tr}[(K_\Gamma^{-1} \Sigma_D)^2] + 2 m_z^* K_\Gamma^{-1} \Sigma_D K_\Gamma^{-1} m_z. \end{aligned}$$

A lognormal moment match gives the reported mean and standard deviation of the LSM score $-\frac{1}{2} \log q_z$ and hence posterior occupancy probabilities.

5 Solver laws and finite-depth qualification

For $0 < m \leq \lambda(A_D) \leq L$ and $\kappa = L/m$, optimal Richardson has

$$\rho_R = \frac{\kappa - 1}{\kappa + 1}, \qquad T_R(\varepsilon) = O\big(\kappa \log \varepsilon^{-1}\big).$$

HB uses

$$\alpha = \frac{4}{(\sqrt{L} + \sqrt{m})^2}, \qquad \beta = \left(\frac{\sqrt{L} - \sqrt{m}}{\sqrt{L} + \sqrt{m}} \right)^2, \quad \rho_H = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1},$$

and Chebyshev semi-iteration has the same asymptotic depth law $O(\sqrt{\kappa} \log \varepsilon^{-1})$. A finite-horizon HB bound contains a polynomial prefactor, typically $(T+1)\rho_H^T$, because endpoint modes have repeated characteristic roots. Exact endpoints may therefore increase the residual before the asymptotic regime. PCG obeys the familiar energy-norm bound $2\rho_H^T$ and terminates in at most n_r steps in exact arithmetic, although finite precision and the large block of probe right-hand sides delay that ideal behavior.

Table 1: Macro-average performance across all scenarios and seeds. Lower is better for residuals and balanced Brier; higher is better otherwise.

Method	AP	area-IoU	mean res.	covariance res.	Brier
learned-Richardson	0.407	0.274	5.817e-01	7.972e-02	0.230
learned-HB	0.784	0.569	3.490e-01	1.420e-01	0.062
learned-Chebyshev	0.785	0.568	3.372e-01	3.295e-02	0.062
global-safe-HB	0.758	0.561	5.583e-01	8.136e-01	0.092
population-safe-HB	0.762	0.544	4.869e-01	7.095e-01	0.074
spectrum-HB	0.225	0.139	4.677e+00	1.317e+01	0.314
population-PCG	0.781	0.554	1.025e-01	4.739e-03	0.090
exact	0.780	0.554	6.802e-05	1.786e-06	0.090

Table 2: Average precision by physical scenario.

Scenario	learned HB	learned Cheb.	population PCG	exact
one obstacle	0.992	0.992	0.975	0.965
two obstacles	0.902	0.901	0.943	0.941
four obstacles	0.784	0.779	0.781	0.779
six obstacles	0.734	0.739	0.725	0.727
three stars	0.823	0.818	0.794	0.796
30 percent noise	0.680	0.683	0.677	0.677
180 degree aperture	0.461	0.470	0.496	0.496
wavenumber 12	0.899	0.898	0.858	0.858

Theorem 1 (Posterior-error propagation). *Let $R_\mu = \Phi - H_D\widehat{Q}_\mu$ and $R_\Sigma = N_D K_\Gamma - H_D\widehat{Q}_\Sigma$. Then*

$$\|\widehat{M}_D - M_D\| \le \|K_\Gamma N_D^*\|\,\|H_D^{-1}\|\,\|R_\mu\|,$$
$$\|\widehat{\Sigma}_D - \Sigma_D\| \le \|K_\Gamma N_D^*\|\,\|H_D^{-1}\|\,\|R_\Sigma\|.$$

Thus a small reconstruction loss alone does not certify Bayesian UQ: the covariance right-hand side must also converge.

Proof. Subtract each approximate equation from (1), multiply by H_D^{-1}, and apply submultiplicativity. □

With n_x probes, a dense recurrent layer costs $O(n_r^2(n_x+n_s))$ after construction of H_D; memory is $O(n_r(n_x+n_s))$. The posterior decoder costs $O(n_s n_r n_x + n_s^2 n_r)$. The spectral sketch dimension is $2(J+1)(2r)^2+4 = 2596$ for $J=8, r=6$, independent of n_x after sketching.

6 Why the loss was not small

There are three distinct obstructions. First, whitening by a small noise scale makes $N_D K_\Gamma N_D^*$ much larger than the identity floor, so typical κ_D is large. Second, a geometry-only population factor has fixed eigendirections and cannot diagonalize every obstacle-dependent H_D. Third, an asymptotically optimal HB pair can have a severe finite-depth transient. The learned controller therefore has a real signal—RHS-weighted spectral statistics predict useful damping—but its best finite-depth output need not enclose the whole spectrum. In the audit, the strict certificate rate is 0.000 for learned HB and 0.906 for learned Chebyshev. Certification and finite-horizon optimality are different goals.

7 Numerical audit

We use $n_r = n_s = 24$, a 28×28 sampling grid, one to six components, frequencies 6–12, 3–30% noise, full and 180-degree apertures, and angular jitter. Training contains one to four disks, ellipses and kites; six components, stars, 30% noise, half aperture and frequency 12 are held out. There are 1 independent seeds, 120 training steps per stationary solver and 4 test tasks per seed and scenario. All principal comparisons use depth $T = 32$. "Spectrum-HB" substitutes the exact eigenvalue endpoints and is an evaluation-only diagnostic. "Exact" uses a dense direct solve.

The macro mean residuals are 3.490e-01 (learned HB), 3.372e-01 (learned Chebyshev), 1.025e-01 (population PCG) and 6.802e-05 (direct). PCG is still 1506 times above the floating-point direct residual at the reported depth,

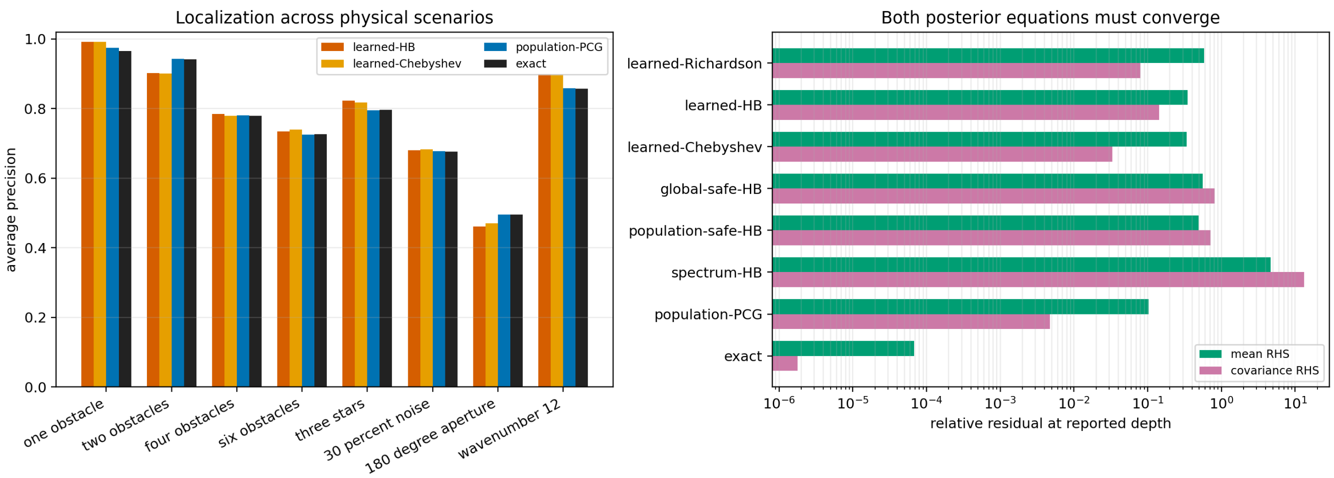

Figure 1: Localization and the two distinct posterior residuals. The exact spectral HB diagnostic exposes the finite-depth transient.

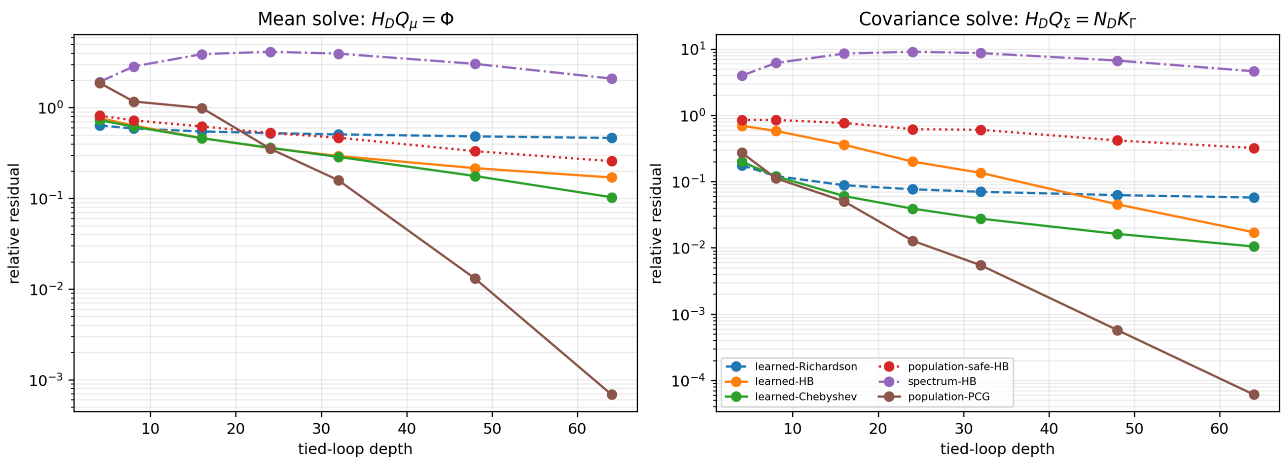

Figure 2: Empirical depth laws on four-obstacle tasks. PCG eventually enters the small-residual regime; stationary methods remain condition-number limited.

so “small” must be defined relative to the inferential target, not only to the direct linear algebra reference. Batched runtimes are 5.78 ms for learned HB, 6.85 ms for learned Chebyshev, 8.94 ms for PCG and 1.82 ms for the small 24×24 dense reference. At this matrix size the direct solve is faster; recurrent scaling becomes relevant only when structure, streaming, hardware reuse or much larger systems prohibit factorization.

8 Interpretation and modelling recommendations

The experiments support five conclusions.

1. The task is genuinely two-dimensional original near-field LSM: the input is a complex multistatic response to several deterministic point sources.

2. The fixed softmax covariance is useful as prior geometry, but choosing it cannot by itself erase the spectrum induced by the current obstacle.

3. Posterior mean and variance must be propagated together. Solving only $H_D Q_\mu = \Phi$ can produce attractive reconstructions with incorrect UQ.

4. Spectral endpoints contain signal, but exact asymptotic HB endpoints are not the finite-depth optimum. The controller should be trained against the two residuals and audited against, rather than forced to equal, the spectrum.

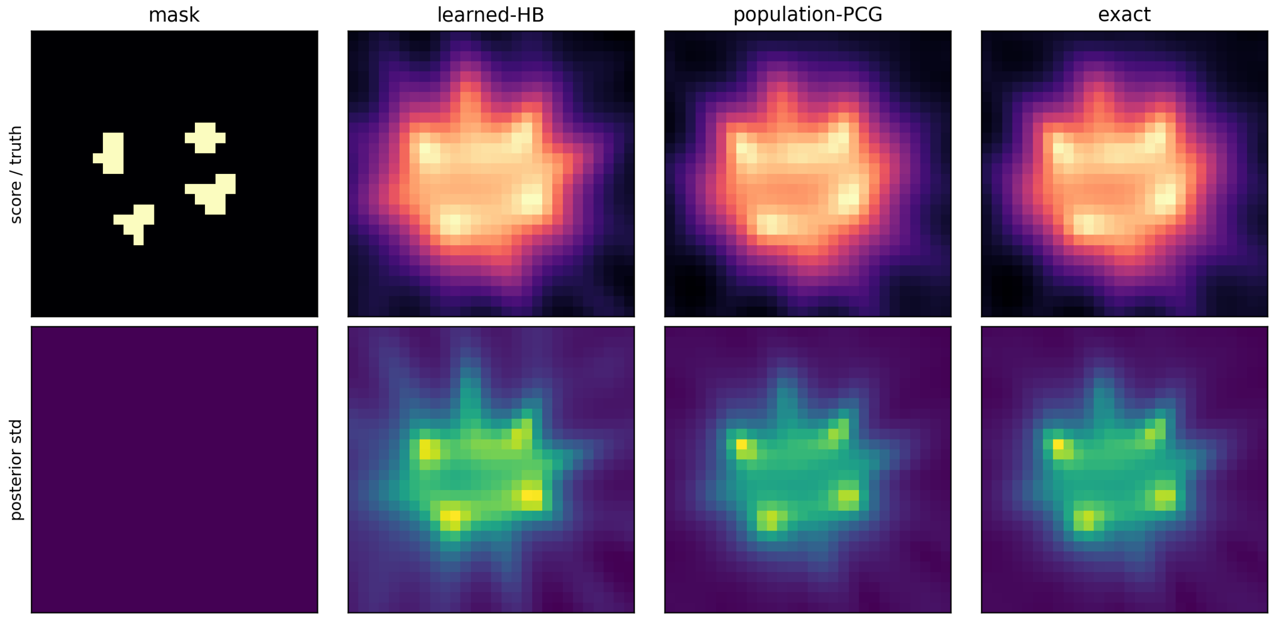

Figure 3: A held-out four-obstacle example. The first row shows truth or posterior score and the second row shows posterior score standard deviation.

5. For a genuinely small loss, use PCG or an A_D-equivariant polynomial preconditioner and increase depth. A larger generic MLP does not repair missing task eigendirections.

The next scientifically necessary step is to replace the MFS generator by an independent high-order boundary-element solver and to test experimental near- field data. Until then, this is a controlled numerical validation, not a claim of field-data performance.

9 Reproducibility

The code fixes all train/evaluation seeds, stores task-level tables and checkpoints, and reports the exact scenario definitions in `protocol.json`. The principal model has 733,731 trainable parameters. Source code and result tables accompany this PDF.

References

[1] G. Kang, C. J. Lee, and X. Cheng. Transformers can learn posterior predictive distributions in-context. *arXiv:2605.26713*, 2026.

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[6] B. T. Polyak. Some methods of speeding up the convergence of iteration methods. *USSR Computational Mathematics and Mathematical Physics*, 1964.

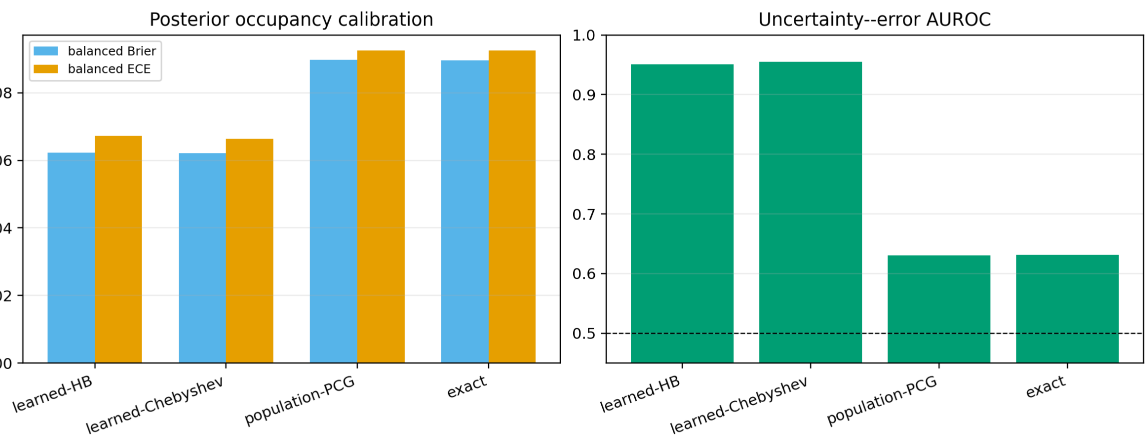

Figure 4: Bayesian UQ obtained from the second parallel recurrence. Thresholds are calibrated separately on training-distribution validation tasks.